

Lecture 3: Review of Probability Theory III

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In this lecture¹, we will introduce pairs of random variables, discussing independence, correlation and covariance, conditional probability and conditional expectation, as well as law of large numbers and central limit theorem.

1 Pairs of R.V.'s

Consider a random experiment with sample space Ω and event class $\mathcal{F}$. We now define a function that assigns to each outcome $\omega \in \Omega$ an ordered pair of real numbers $(X(\omega), Y(\omega))$. In other words, we are dealing with a vector-valued R.V. that maps Ω into the real plane $\mathbb{R}^2$.

Suppose (X, Y) takes values from a countable set $\Omega_{X,Y} = \{(x_j, y_k) : j = 1, 2, \dots; k = 1, 2, \dots\}$. The **joint probability mass function** of (X, Y) specifies the probability of the event $X = x, Y = y$ for each pair $(x, y) \in \Omega_{X,Y}$.

$$P_{X,Y}(x, y) = P[\{X = x\} \cap \{Y = y\}] \triangleq P[X = x, Y = y]$$

for $x, y \in \mathbb{R}^2$.

The joint PMF of (X, Y) characterizes the joint behavior of X and Y . We are also interested in the probabilities of events involving each R.V. individually. These can be obtained from the marginal PMFs:

$$p_X(x_j) = P(X = x_j) = P(X = x_j, Y = \text{Anything}) = \sum_{k=1}^{\infty} P(X = x_j, Y = y_k),$$

similarly,

$$p_Y(y_k) = P(Y = y_k) = P(X = \text{Anything}, Y = y_k) = \sum_{j=1}^{\infty} P(X = x_j, Y = y_k),$$

The **joint cumulative distribution function** of X and Y is defined as the probability of the event $\{X \leq x_1\} \cap \{Y \leq y_1\}$:

$$F_{X,Y}(x_1, y_1) = P(X \leq x_1, Y \leq y_1).$$

The joint CDF satisfies the following properties.

(i) **Monotonicity.** The joint cdf is a nondecreasing function of x and y :

$$F_{X,Y}(x_1, y_1) \leq F_{X,Y}(x_2, y_2) \quad \text{if } x_1 \leq x_2 \text{ and } y_1 \leq y_2. \quad (1)$$

(ii) **Boundary values.**

$$F_{X,Y}(x_1, -\infty) = 0, \quad F_{X,Y}(-\infty, y_1) = 0, \quad F_{X,Y}(\infty, \infty) = 1. \quad (2)$$

(iii) **Marginals (remove one constraint).**

$$F_X(x_1) = F_{X,Y}(x_1, \infty) \quad \text{and} \quad F_Y(y_1) = F_{X,Y}(\infty, y_1). \quad (3)$$

¹Reading: Chapter 5-7 of Leon-Garcia.

(iv) **Right-continuity in each coordinate (“north” and “east”).**

$$\lim_{x \rightarrow a^+} F_{X,Y}(x, y) = F_{X,Y}(a, y) \quad \text{and} \quad \lim_{y \rightarrow b^+} F_{X,Y}(x, y) = F_{X,Y}(x, b). \quad (4)$$

(v) **Probability of a rectangle.**

$$P(x_1 < X \leq x_2, y_1 < Y \leq y_2) = F_{X,Y}(x_2, y_2) - F_{X,Y}(x_2, y_1) - F_{X,Y}(x_1, y_2) + F_{X,Y}(x_1, y_1). \quad (5)$$

We say that the R.V.'s X and Y are jointly continuous if the probabilities of events involving (X, Y) can be expressed as an integral of a PDF. Specifically, there exists a nonnegative function $f_{X,Y}(x, y)$ called the joint PDF, defined on the real plane such that for every event B , a subset of the plane,

$$P((X, Y) \in B) = \int_B \int f_{X,Y}(x', y') dx' dy'.$$

When B is the entire plane, the integral must equal one:

$$\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f_{X,Y}(x', y') dx' dy' = 1.$$

The joint CDF can be obtained in terms of the joint PDF of jointly continuous R.V.'s by integrating over the semi-infinite rectangle defined by (x, y) :

$$F_{X,Y}(x, y) = \int_{-\infty}^x \int_{-\infty}^y f_{X,Y}(x', y') dx' dy'.$$

It then follows that if X and Y are jointly continuous R.V.'s, then the PDF can be obtained from the CDF by differentiation:

$$f_{X,Y}(x', y') = \frac{\partial F_{X,Y}(x, y)}{\partial x \partial y}.$$

The probability of a rectangular region is obtained by letting $B = \{(x, y) : a_1 < x \leq b_1, a_2 < y \leq b_2\}$:

$$P(a_1 < X \leq b_1, a_2 < Y \leq b_2) = \int_{a_1}^{b_1} \int_{a_2}^{b_2} f_{X,Y}(x', y') dx' dy'.$$

The marginal PDF's $f_X(x)$ and $f_Y(y)$ are obtained by taking the derivative of the corresponding marginal CDF's, $F_X(x) = F_{X,Y}(x, \infty)$ and $F_Y(y) = F_{X,Y}(\infty, y)$. Thus,

$$f_X(x) = \int_{-\infty}^{\infty} f_{X,Y}(x, y') dy',$$

and

$$f_Y(y) = \int_{-\infty}^{\infty} f_{X,Y}(x', y) dx'.$$

2 Independence, Conditional Probability and Conditional Expectation

Discrete R.V.'s X and Y are independent if and only if the joint PMF is equal to the product of the marginal PMF's

$$p_{X,Y}(x_i, y_k) = P(X = x_j, Y = y_k) = P(X = x_j)P(Y = y_k) = p_X(x_j)p_Y(y_k).$$

In general, two R.V.'s X and Y are independent if and only if their joint CDF equals the product of their marginal CDFs.

$$F_{X,Y}(x,y) = F_X(x)F_Y(y) \quad \text{for } \forall x,y$$

Similarly, if X and Y are jointly continuous, then they are independent if and only if their joint PDF equals the product of their marginal PDFs:

$$f_{X,Y}(x,y) = f_X(x)f_Y(y) \quad \text{for } \forall x,y$$

Remark 2.1. If X and Y are independent R.V.'s, then the R.V.'s defined by any pair of functions $g(X)$ and $h(Y)$ are also independent.

The joint moments of two R.V. X and Y summarize information about their joint behavior. The jk th joint moment of X and Y is defined by

$$\mathbb{E}[X^j Y^k] = \begin{cases} \iint_{-\infty}^{\infty} x^j y^k f_{X,Y}(x,y) dx dy, & X,Y \text{ jointly continuous,} \\ \sum_i \sum_n x_i^j y_n^k p_{X,Y}(x_i, y_n), & X,Y \text{ discrete.} \end{cases}$$

It is customary to refer to the $j = 1, k = 1$ moment, $\mathbb{E}[XY]$, as the correlation of X and Y . When $\mathbb{E}[XY] = 0$, we say that X and Y are orthogonal.

The jk th central moment of X and Y is defined as the joint moment of the centered R.V.'s $X - \mathbb{E}[X]$ and $Y - \mathbb{E}[Y]$,

$$\mathbb{E}[(X - \mathbb{E}[X])^j (Y - \mathbb{E}[Y])^k].$$

The covariance of X and Y is defined as the $j = k = 1$ central moment:

$$\text{Cov}(X, Y) \triangleq \mathbb{E}[(X - \mathbb{E}[X])(Y - \mathbb{E}[Y])] = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y].$$

Note that $\text{Cov}(X, Y) = \mathbb{E}[XY]$ if either of the random variables has mean zero.

Example 1. Let X and Y be independent random variables. Find their covariance.

$$\text{Cov}(X, Y) = \mathbb{E}[(X - \mathbb{E}[X])(Y - \mathbb{E}[Y])] = \mathbb{E}[X - \mathbb{E}[X]] \mathbb{E}[Y - \mathbb{E}[Y]] = 0,$$

where the second equality follows from the fact that X and Y are independent, and the third equality follows from

$$\mathbb{E}[X - \mathbb{E}[X]] = \mathbb{E}[X] - \mathbb{E}[X] = 0.$$

Therefore *pairs of independent random variables have covariance zero.*

Multiplying either X or Y by a large constant increases the covariance, so we normalize it in order to measure correlation on an absolute scale.

Definition 2.1 (Correlation Coefficient). The correlation coefficient of X and Y is defined by

$$\rho_{X,Y} = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y} = \frac{\mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y]}{\sigma_X \sigma_Y},$$

where $\sigma_X = \sqrt{\text{Var}(X)}$ and $\sigma_Y = \sqrt{\text{Var}(Y)}$ are the standard deviations of X and Y , respectively.

The correlation coefficient is a number that is at most 1 in magnitude:

$$-1 \leq \rho_{X,Y} \leq 1. \quad (6)$$

To establish the inequalities, we start from the fact that the expected square of a R.V. is nonnegative, which yields the following inequality:

$$\begin{aligned} 0 &\leq \mathbb{E} \left\{ \left(\frac{X - \mathbb{E}[X]}{\sigma_X} \pm \frac{Y - \mathbb{E}[Y]}{\sigma_Y} \right)^2 \right\} \\ &= 1 \pm 2\rho_{X,Y} + 1 \\ &= 2(1 \pm \rho_{X,Y}). \end{aligned}$$

The last equation implies Eq. (6).

The extreme values of $\rho_{X,Y}$ are achieved when X and Y are related linearly, $Y = aX + b$; $\rho_{X,Y} = 1$ if $a > 0$ and $\rho_{X,Y} = -1$ if $a < 0$.

R.V.'s X and Y are said to be **uncorrelated** if $\rho_{X,Y} = 0$. If X and Y are independent, then $\text{Cov}(X, Y) = 0$, so $\rho_{X,Y} = 0$. Thus if X and Y are independent, then X and Y are uncorrelated. The following example shows that **it is possible for X and Y to be uncorrelated but not independent**.

Example 2. Let Θ be uniformly distributed in the interval $(0, 2\pi)$. Let

$$X = \cos \Theta \quad \text{and} \quad Y = \sin \Theta.$$

The point (X, Y) then corresponds to the point on the unit circle specified by the angle Θ . The marginal PDF's of X and Y are arcsine PDF's, which are nonzero in the interval $(-1, 1)$. The product of the marginals is nonzero in the square defined by $-1 \leq x \leq 1$ and $-1 \leq y \leq 1$, so if X and Y were independent the point (X, Y) would assume all values in this square. This is not the case, so X and Y are dependent.

We now show that X and Y are uncorrelated:

$$\begin{aligned} \mathbb{E}[XY] &= \mathbb{E}[\sin \Theta \cos \Theta] = \frac{1}{2\pi} \int_0^{2\pi} \sin \phi \cos \phi \, d\phi \\ &= \frac{1}{4\pi} \int_0^{2\pi} \sin 2\phi \, d\phi = 0. \end{aligned}$$

Since $\mathbb{E}[X] = \mathbb{E}[Y] = 0$, then it implies that X and Y are uncorrelated.

Note that if X and Y are jointly Gaussian and $\rho_{X,Y} = 0$ then X and Y are independent random variables.

Definition 2.2 (Conditional PMF). For X and Y discrete random variables, the conditional PMF of Y given $X = x$ is defined by:

$$p_Y(y | x) = P(Y = y | X = x) = \frac{P(X = x, Y = y)}{P(X = x)} = \frac{p_{X,Y}(x, y)}{p_X(x)}$$

for x such that $P(X = x) > 0$.

We define $p_Y(y | x) = 0$ for x s.t. $P(X = x) = 0$. Note that $p_Y(y | x)$ is a function of y over the real line, and that $p_Y(y | x) > 0$ only for y in a discrete set $\{y_1, y_2, \dots\}$.

Definition 2.3 (Conditional CDF). Suppose Y is a continuous random variable and X is a discrete R.V., then the conditional CDF of Y given $X = x_k$ is defined as

$$F_Y(y | x_k) = \frac{P(Y \leq y, X = x_k)}{P(X = x_k)}, \quad \text{for } P(X = x_k) > 0.$$

If X is a continuous R.V., then the conditional CDF of Y given $X = x$ is defined as

$$F_Y(y | x) = \lim_{h \rightarrow 0} F_Y(y | x < X \leq x + h).$$

Notice that

$$\begin{aligned}
F_Y(y \mid x < X \leq x+h) &= \frac{P[Y \leq y, x < X \leq x+h]}{P[x < X \leq x+h]} \\
&= \frac{\int_{-\infty}^y \int_x^{x+h} f_{X,Y}(x', y') dx' dy'}{\int_x^{x+h} f_X(x') dx'} \\
&= \frac{\int_{-\infty}^y f_{X,Y}(x, y') dy' h}{f_X(x)h}.
\end{aligned}$$

As we let h approach zero,

$$F_Y(y \mid x) = \frac{\int_{-\infty}^y f_{X,Y}(x, y') dy'}{f_X(x)}. \quad (7)$$

Definition 2.4 (Conditional PDF). The conditional PDF of Y given $X = x_k$, if the derivative exists, is given by

$$f_Y(y \mid x_k) = \frac{d}{dy} F_Y(y \mid x_k),$$

and the conditional PDF of Y given continuous $X = x$ is then:

$$f_Y(y \mid x) = \frac{d}{dy} F_Y(y \mid x) = \frac{f_{X,Y}(x, y)}{f_X(x)}. \quad (8)$$

Definition 2.5 (Conditional Expectation). The conditional expectation of Y given $X = x$ is defined by

$$\mathbb{E}[Y \mid x] = \int_{-\infty}^{\infty} y f_Y(y \mid x) dy.$$

In the special case where X and Y are both discrete random variables we have:

$$\mathbb{E}[Y \mid x_k] = \sum_{y_j} y_j p_Y(y_j \mid x_k). \quad (9)$$

Clearly, $\mathbb{E}[Y \mid x]$ is simply the center of mass associated with the conditional PDF or PMF.

Remark 2.2. The conditional expectation $\mathbb{E}[Y \mid x]$ can be viewed as defining a function of x : $g(x) = \mathbb{E}[Y \mid x]$. It therefore makes sense to talk about the random variable $g(X) = \mathbb{E}[Y \mid X]$. We can imagine that a random experiment is performed and a value for X is obtained, say $X = x_0$, and then the value $g(x_0) = \mathbb{E}[Y \mid x_0]$ is produced. We are interested in $\mathbb{E}[g(X)] = \mathbb{E}[\mathbb{E}[Y \mid X]]$. In particular, we have

$$\mathbb{E}[Y] = \mathbb{E}[\mathbb{E}[Y \mid X]], \quad (10)$$

where the RHS is

$$\mathbb{E}[\mathbb{E}[Y \mid X]] = \int_{-\infty}^{\infty} \mathbb{E}[Y \mid x] f_X(x) dx, \quad (11)$$

$$\mathbb{E}[\mathbb{E}[Y \mid X]] = \sum_{x_k} \mathbb{E}[Y \mid x_k] p_X(x_k) \quad (12)$$

If X and Y are jointly continuous random variables. Then

$$\begin{aligned}
\mathbb{E}[\mathbb{E}[Y | X]] &= \int_{-\infty}^{\infty} \mathbb{E}[Y | x] f_X(x) dx \\
&= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} y f_Y(y | x) dy f_X(x) dx \\
&= \int_{-\infty}^{\infty} y \int_{-\infty}^{\infty} f_{X,Y}(x, y) dx dy \\
&= \int_{-\infty}^{\infty} y f_Y(y) dy \\
&= \mathbb{E}[Y].
\end{aligned}$$

The above result also holds for the expected value of a function of Y :

$$\mathbb{E}[h(Y)] = \mathbb{E}[\mathbb{E}[h(Y) | X]].$$

In particular, the k th moment of Y is given by

$$\mathbb{E}[Y^k] = \mathbb{E}[\mathbb{E}[Y^k | X]].$$

Conditional expectation as an optimal estimator. Suppose that Y is real-valued and $\mathbb{E}[Y^2] < \infty$. After observing X , we may estimate Y by a measurable function $h(X)$. The conditional mean $g(X) = \mathbb{E}[Y | X]$ is the *minimum mean-square error (MMSE) estimator*: for every such estimator with $\mathbb{E}[h(X)^2] < \infty$,

$$\mathbb{E}[(Y - h(X))^2] = \mathbb{E}[(Y - g(X))^2] + \mathbb{E}[(g(X) - h(X))^2] \geq \mathbb{E}[(Y - g(X))^2].$$

Thus, the optimal estimate of Y from an observed value $X = x$ is $g(x) = \mathbb{E}[Y | X = x]$ under squared-error loss. The minimizing estimator is unique up to almost-sure equality.

Proof sketch. Write $Y - h(X) = (Y - g(X)) + (g(X) - h(X))$ and expand the square. The cross term vanishes by conditioning on X :

$$\mathbb{E}[(Y - g(X))(g(X) - h(X))] = \mathbb{E}[(g(X) - h(X))\mathbb{E}[Y - g(X) | X]] = 0.$$

This gives the stated decomposition. Equality holds if and only if $h(X) = g(X)$ almost surely.

Geometric interpretation. The estimation error $Y - \mathbb{E}[Y | X]$ is orthogonal to every square-integrable function of the observed variable X .

3 Multiple R.V.'s

Random vector $\mathbf{X} = [X_1, \dots, X_n]$

Definition 3.1 (Joint CDF). The joint CDF of $X_1, \dots, X_n$ is defined as the probability of an n -dimensional semi-infinite rectangle associated with the point $(x_1, \dots, x_n)$:

$$F_{\mathbf{X}}(\mathbf{x}) \triangleq F_{X_1, \dots, X_n}(x_1, \dots, x_n) = P(X_1 \leq x_1, X_2 \leq x_2, \dots, X_n \leq x_n).$$

Definition 3.2 (Joint PMF). The joint PMF of n discrete random variables is defined by

$$p_{\mathbf{X}}(\mathbf{x}) \triangleq p_{X_1, \dots, X_n}(x_1, \dots, x_n) = P(X_1 = x_1, X_2 = x_2, \dots, X_n = x_n).$$

A family of **conditional PMF's** is obtained from the joint PMF by conditioning on different subcollections of the random variables. For example, if $p_{X_1, \dots, X_{n-1}}(x_1, \dots, x_{n-1}) > 0$, then

$$p_{X_n}(x_n | x_1, \dots, x_{n-1}) = \frac{p_{X_1, \dots, X_n}(x_1, \dots, x_n)}{p_{X_1, \dots, X_{n-1}}(x_1, \dots, x_{n-1})}.$$

Repeated applications yield the following very useful expression (chain rule):

$$p_{X_1, \dots, X_n}(x_1, \dots, x_n) = p_{X_n}(x_n | x_1, \dots, x_{n-1}) p_{X_{n-1}}(x_{n-1} | x_1, \dots, x_{n-2}) \cdots p_{X_2}(x_2 | x_1) p_{X_1}(x_1).$$

Random variables $X_1, X_2, \dots, X_n$ are **jointly continuous random variables** if the probability of any n -dimensional event A is given by an n -dimensional integral of a probability density function:

$$P(\mathbf{X} \in A) = \int_{\mathbf{x} \in A} \int \cdots \int f_{X_1, \dots, X_n}(x'_1, \dots, x'_n) dx'_1 \cdots dx'_n,$$

where $f_{X_1, \dots, X_n}(x_1, \dots, x_n)$ is the **joint PDF**.

The joint CDF of $\mathbf{X}$ is obtained from the joint PDF by integration:

$$F_{\mathbf{X}}(\mathbf{x}) = F_{X_1, \dots, X_n}(x_1, \dots, x_n) = \int_{-\infty}^{x_1} \cdots \int_{-\infty}^{x_n} f_{X_1, \dots, X_n}(x'_1, \dots, x'_n) dx'_1 \cdots dx'_n.$$

The joint PDF (if the derivative exists) is given by

$$f_{\mathbf{X}}(\mathbf{x}) \triangleq f_{X_1, \dots, X_n}(x_1, \dots, x_n) = \frac{\partial^n}{\partial x_1 \cdots \partial x_n} F_{X_1, \dots, X_n}(x_1, \dots, x_n).$$

The marginal PDF for a subset of the random variables is obtained by integrating the other variables out. For example, the marginal pdf of X_1 is

$$f_{X_1}(x_1) = \int_{-\infty}^{\infty} \cdots \int_{-\infty}^{\infty} f_{X_1, \dots, X_n}(x_1, x'_2, \dots, x'_n) dx'_2 \cdots dx'_n.$$

As another example, the marginal PDF for $X_1, \dots, X_{n-1}$ is given by

$$f_{X_1, \dots, X_{n-1}}(x_1, \dots, x_{n-1}) = \int_{-\infty}^{\infty} f_{X_1, \dots, X_n}(x_1, \dots, x_{n-1}, x'_n) dx'_n.$$

A family of **conditional PDF's** is also associated with the joint pdf. For example, the PDF of X_n given the values of $X_1, \dots, X_{n-1}$ is

$$f_{X_n}(x_n | x_1, \dots, x_{n-1}) = \frac{f_{X_1, \dots, X_n}(x_1, \dots, x_n)}{f_{X_1, \dots, X_{n-1}}(x_1, \dots, x_{n-1})}.$$

if $f_{X_1, \dots, X_{n-1}}(x_1, \dots, x_{n-1}) > 0$.

Repeated such applications yield an expression:

$$f_{X_1, \dots, X_n}(x_1, \dots, x_n) = f_{X_n}(x_n | x_1, \dots, x_{n-1}) f_{X_{n-1}}(x_{n-1} | x_1, \dots, x_{n-2}) \cdots f_{X_2}(x_2 | x_1) f_{X_1}(x_1).$$

Independence As shown in Table 1, independence means the joint distribution factorizes into the product of the marginals.

Case	Independence Condition
General (CDF)	$F_{X_1, \dots, X_n}(x_1, \dots, x_n) = F_{X_1}(x_1) \cdots F_{X_n}(x_n)$
Discrete (PMF)	$p_{X_1, \dots, X_n}(x_1, \dots, x_n) = p_{X_1}(x_1) \cdots p_{X_n}(x_n)$
Continuous (PDF)	$f_{X_1, \dots, X_n}(x_1, \dots, x_n) = f_{X_1}(x_1) \cdots f_{X_n}(x_n)$

Table 1: Equivalent formulations of independence for random variables

4 Functions of Several R.V.'s and Their Expected Value

For $\mathbf{X} = (X_1, X_2, \dots, X_n)$, the **mean vector** is defined as the column vector of expected values of the components X_k :

$$\mathbf{m}_X = \mathbb{E}[\mathbf{X}] = \mathbb{E} \begin{bmatrix} X_1 \\ X_2 \\ \vdots \\ X_n \end{bmatrix} \triangleq \begin{bmatrix} \mathbb{E}[X_1] \\ \mathbb{E}[X_2] \\ \vdots \\ \mathbb{E}[X_n] \end{bmatrix}.$$

The **correlation matrix** has the second moments of $\mathbf{X}$ as its entries:

$$\mathbf{R}_X = \begin{bmatrix} \mathbb{E}[X_1^2] & \mathbb{E}[X_1 X_2] & \cdots & \mathbb{E}[X_1 X_n] \\ \mathbb{E}[X_2 X_1] & \mathbb{E}[X_2^2] & \cdots & \mathbb{E}[X_2 X_n] \\ \vdots & \vdots & \ddots & \vdots \\ \mathbb{E}[X_n X_1] & \mathbb{E}[X_n X_2] & \cdots & \mathbb{E}[X_n^2] \end{bmatrix}.$$

The **covariance matrix** of $\mathbf{X} = (X_1, X_2, \dots, X_n)$ is defined as

$$\text{Cov}(\mathbf{X}) = \mathbb{E}[(\mathbf{X} - \mathbf{m}_X)(\mathbf{X} - \mathbf{m}_X)^\top].$$

In expanded form,

$$\mathbf{K}_X = \text{Cov}(\mathbf{X}) = \begin{bmatrix} \text{Var}(X_1) & \text{Cov}(X_1, X_2) & \cdots & \text{Cov}(X_1, X_n) \\ \text{Cov}(X_2, X_1) & \text{Var}(X_2) & \cdots & \text{Cov}(X_2, X_n) \\ \vdots & \vdots & \ddots & \vdots \\ \text{Cov}(X_n, X_1) & \text{Cov}(X_n, X_2) & \cdots & \text{Var}(X_n) \end{bmatrix}.$$

Both $\mathbf{R}_X$ and $\mathbf{K}_X$ are $n \times n$ symmetric matrices. The diagonal elements of $\mathbf{K}_X$ are given by the variances $\text{Var}[X_k] = \mathbb{E}[(X_k - m_k)^2]$ of the components of $\mathbf{X}$. If these elements are uncorrelated, then $\text{Cov}(X_j, X_k) = 0$ for $j \neq k$, and $\mathbf{K}_X$ is diagonal. If the random variables $X_1, \dots, X_n$ are independent, then they are uncorrelated and $\mathbf{K}_X$ is diagonal. Finally, if the vector of expected values is $\mathbf{0}$, that is, $m_k = \mathbb{E}[X_k] = 0$ for all k , then

$$\mathbf{R}_X = \mathbf{K}_X.$$

5 Two Limit Theorems: LLN and CLT

Let $X_1, X_2, \dots$ be independent and identically distributed (i.i.d.) random variables with mean m and finite variance σ^2 . Define the sum and the sample mean by

$$S_n = \sum_{i=1}^n X_i, \quad \bar{X}_n = \frac{S_n}{n}.$$

The law of large numbers explains why averages stabilize; the central limit theorem describes their fluctuations.

5.1 The law of large numbers: why averages stabilize

Theorem 5.1 (Weak law of large numbers: finite-variance version). *Under the above assumptions, for every $\varepsilon > 0$,*

$$\mathbb{P}(|\bar{X}_n - m| \geq \varepsilon) \leq \frac{\sigma^2}{n\varepsilon^2} \rightarrow 0.$$

Thus $\bar{X}_n$ converges in probability to m , written $\bar{X}_n \xrightarrow{\mathbb{P}} m$.

Proof. Linearity of expectation and independence give

$$\mathbb{E}[\bar{X}_n] = m, \quad \text{Var}(\bar{X}_n) = \frac{1}{n^2} \sum_{i=1}^n \text{Var}(X_i) = \frac{\sigma^2}{n}.$$

Apply Chebyshev's inequality to $\bar{X}_n$ with threshold ε and let $n \rightarrow \infty$. □

Remark 5.1 (What the weak law does—and does not—say). *For any fixed tolerance, the probability of a large error vanishes as the sample size increases. It does not say that the average becomes exactly m at a finite time, or that each additional observation brings it closer to m . The mechanism is that the variance of the average decreases as $1/n$, not that future observations compensate for past outcomes.*

Theorem 5.2 (Strong law of large numbers). *More generally, if $X_1, X_2, \dots$ are i.i.d. and $\mathbb{E}[|X_1|] < \infty$, then*

$$\mathbb{P}\left(\lim_{n \rightarrow \infty} \bar{X}_n = \mathbb{E}[X_1]\right) = 1.$$

This is almost-sure convergence, written $\bar{X}_n \xrightarrow{\text{a.s.}} \mathbb{E}[X_1]$.

The strong law describes convergence along almost every infinite sample sequence and implies the weak law. “Strong” refers to the mode of convergence, not a faster convergence rate. Finite variance is sufficient but not necessary for either law. The strong law is stated without proof.

Example: estimating a probability by simulation. In independent repetitions of an experiment, let $X_i = 1$ if an event occurs and $X_i = 0$ otherwise, with event probability p . Then $\hat{p}_n = \bar{X}_n$ is the observed frequency and

$$\mathbb{P}(|\hat{p}_n - p| \geq \varepsilon) \leq \frac{p(1-p)}{n\varepsilon^2} \leq \frac{1}{4n\varepsilon^2}.$$

For $\varepsilon > 0$ and $0 < \delta < 1$, to guarantee an error smaller than ε with probability at least $1 - \delta$, it suffices to take $n \geq 1/(4\delta\varepsilon^2)$. This is a rigorous finite-sample guarantee, although it can be conservative.

5.2 The central limit theorem: the shape of fluctuations

Theorem 5.3 (Central limit theorem: i.i.d. version). *Let $X_1, X_2, \dots$ be i.i.d. with mean m and variance $0 < \sigma^2 < \infty$. Then*

$$Z_n = \frac{S_n - nm}{\sigma\sqrt{n}} = \frac{\sqrt{n}(\bar{X}_n - m)}{\sigma} \xrightarrow{d} \mathcal{N}(0, 1).$$

Here, convergence in distribution means that, for every $z \in \mathbb{R}$,

$$\lim_{n \rightarrow \infty} \mathbb{P}(Z_n \leq z) = \Phi(z), \quad \Phi(z) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^z e^{-u^2/2} du.$$

The CLT is stated without proof.

Remark 5.2 (LLN versus CLT). *The LLN identifies the limiting average; the CLT describes the distribution of its error after rescaling. The standard deviation of $\bar{X}_n$ is $\sigma/\sqrt{n}$, so multiplying the error by $\sqrt{n}$ keeps its fluctuations visible. Equivalently, the large- n normal approximations are*

$$\bar{X}_n \approx \mathcal{N}\left(m, \frac{\sigma^2}{n}\right), \quad S_n \approx \mathcal{N}(nm, n\sigma^2).$$

Thus, the average becomes more concentrated, even though the standard deviation of the sum grows as $\sqrt{n}$. Quadrupling the sample size halves the standard deviation of the average.

Example continued: a bound versus an approximation. For $p = 1/2$ and $n = 2500$, the sample proportion has standard deviation 0.01. Chebyshev's inequality guarantees

$$\mathbb{P}(|\hat{p}_{2500} - 0.5| < 0.02) \geq 1 - \frac{0.25}{2500(0.02)^2} = 0.75.$$

The CLT instead gives the normal approximation

$$\mathbb{P}(|\hat{p}_{2500} - 0.5| < 0.02) \approx \mathbb{P}(|Z| < 2) = 2\Phi(2) - 1 \approx 0.9545, \quad Z \sim \mathcal{N}(0, 1).$$

The first statement is a guaranteed lower bound; the second is an asymptotic approximation, not a guarantee. The displayed approximation omits a continuity correction.

Remark 5.3 (Common misconceptions and the role of assumptions). *The CLT does not require the individual X_i to be Gaussian, nor does it say that they become Gaussian as n increases. It concerns the standardized sum. The approximation need not be accurate at every finite sample size; there is no universal rule that $n \geq 30$ is sufficient for every distribution.*

Independence also matters. If $X_i = Y$ for every i , where Y equals -1 or 1 with equal probability, then $\bar{X}_n = Y$ for all n : averaging does not reduce uncertainty at all. For dependent observations in a stochastic process, the above i.i.d. theorems cannot be applied automatically; additional assumptions are needed.